Computer and Systems

Department

Faculty of Engineering

Ain Shams University

[CSE211s] Introduction To Embedded

Project Report

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Abstract

This project was undertaken as part of the "Introduction to Embedded Systems" course. Its objective is to design a GPS positioning system capable of acquiring real-time location data via a microcontroller, detecting nearby landmarks, and presenting the information on an LCD display.

Introduction

This project implements a real-time location tracking solution using a GPS module connected to a microcontroller. Powered by the TM4C123G Launchpad, the system gathers positional information on the move, recognizes nearby landmarks from a preset database, and presents their names on an LCD screen. Designed for navigation within the faculty, the system highlights essential embedded development practices, including UART interfacing and real-time data handling.

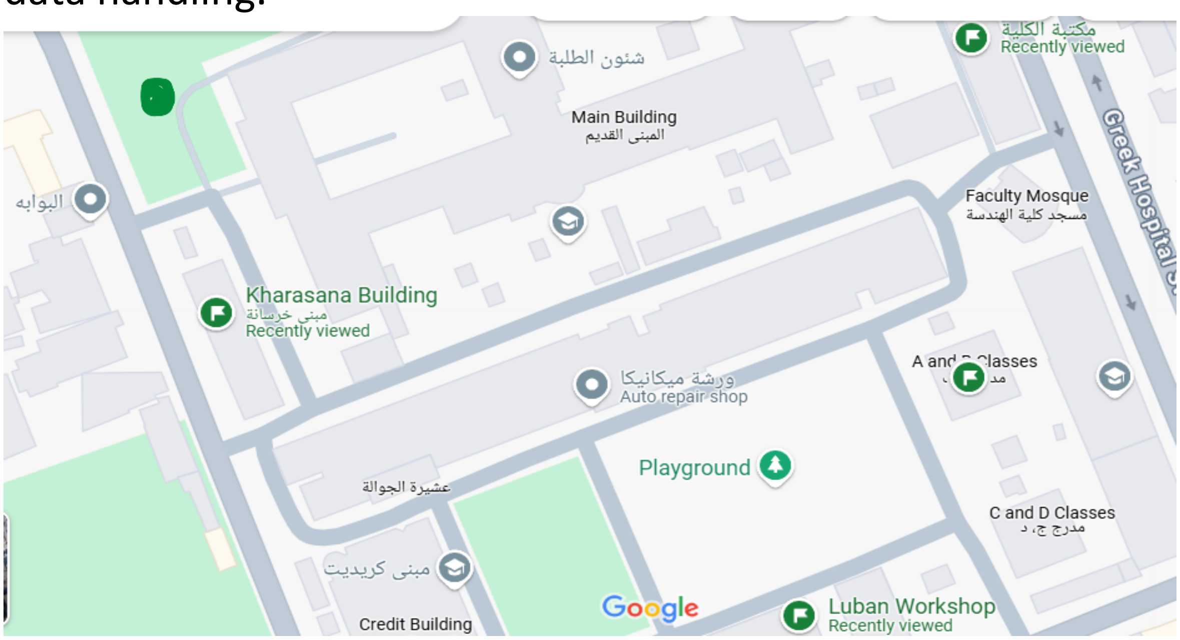

Figure 1: Faculty map with Marked Landmarks

Objectives

- Design an embedded system capable of real-time location tracking through GPS integration.
- Utilize UART protocol to retrieve and decode GPS coordinate data.
- Maintain a database of predefined landmark coordinates within the faculty environment.
- Display the name of the closest landmark on an LCD based on the current position.
- Continuously update the displayed location as the microcontroller moves.

Bounses

1. **Custom-Built LCD Driver:** The LCD functionality was implemented manually, without the use of pre-existing libraries.
2. **Independent power source:** The system runs entirely on external power source.
3. **Live Speed Monitoring:** Computes and displays live speed based on GPS input.
4. **Protective Packaging:** The system is enclosed in a custom-designed case that protects internal components and provides a clean, organized presentation. The case includes cutouts for the LCD, buttons, and ports, allowing full access and portability.
5. **Adjustable LCD Brightness** – A built-in potentiometer allows manual adjustment of the LCD's brightness and contrast. This ensures optimal screen visibility in varying lighting conditions and enhances user experience.

Contribution List

Name	ID	Contribution
Seif Eldin Mustafa Abdel Fattah	2200794	GPS Driver Code - Report
Ammar Ahmed Wahidi	2200360	GPIO - UART - SYSTICK Drivers Code
Omar Ahmed Abd El-Kader	2200706	Main Code
Abdelrahman Essam Fahmy	2200251	Main Code
Taha Seif	2200671	UART Driver Code - Packaging
Sobhey Mohamed Osman	2200477	LCD Driver Code

System Design

The system uses a GPS module interfaced with a microcontroller to determine the user's current position and pinpoint nearby landmarks. GPS data is transmitted via UART and processed by the microcontroller, which compares it to a predefined set of landmark coordinates. The nearest match is then shown on an LCD display. Optimized for speed and low power consumption, this design offers a practical solution for real-time navigation within faculty grounds.

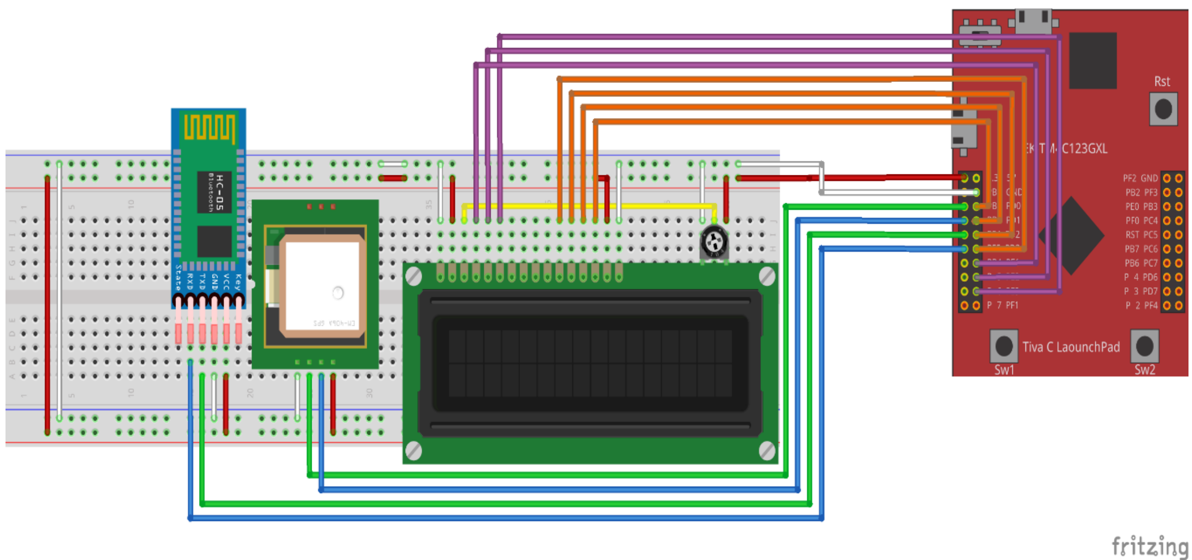

Figure 2: System schematic

Hardware Components

TM4C123G Launchpad (Microcontroller)

Equipped with an ARM Cortex-M4 processor, the TM4C123G Launchpad serves as the central controller in GPS-driven embedded systems. It manages real-time data processing and communication, ensuring smooth operation of system tasks. With built-in features like GPIOs, timers, and UART interfaces, the board enables efficient interaction with GPS components and facilitates reliable coordination across the system.

Figure 3: TM4C123G

GPS Module

The GPS module acquires real-time latitude and longitude data by communicating with satellite networks. This positional information is transmitted to the microcontroller over UART using the NMEA protocol. The microcontroller processes the data to identify nearby landmarks and updates the LCD screen with the corresponding location details.

Figure 4: GPS Module

LCD Display

The LCD screen provides real-time visual feedback by displaying the current GPS coordinates and the names of nearby landmarks. This allows users to monitor location updates directly, without relying on a computer. The microcontroller communicates with the LCD using GPIO pins to send properly formatted output for display.

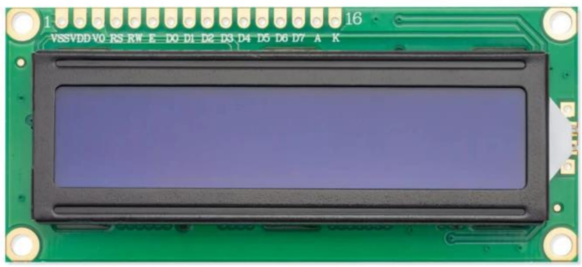

Figure 5: LCD Display

Software Dependencies and Configuration

Keil µVision IDE

Keil µVision is an integrated development environment (IDE) tailored for embedded C programming. It equips developers with essential features such as code compilation, debugging tools, and firmware flashing capabilities for the TM4C123G Launchpad. This setup enables streamlined development and reliable hardware integration in embedded system projects.

Embedded C Programming
This project is developed using Embedded C, a hardware-oriented extension of the C programming language. It allows precise control over peripheral registers, facilitates structured UART communication, and manages the processing of GPS data at a low level.

Docklight (Serial Monitor)
Docklight is a specialized serial communication monitoring and testing tool used to observe and simulate UART data exchange between the microcontroller and a PC. It enables real-time analysis by capturing and displaying incoming GPS data, sending test commands, and logging communication events. Docklight is particularly useful for debugging complex serial protocols, validating response patterns, and ensuring robust communication between system components.

Implementation Strategy

This section outlines the development and operational flow of the system. The implementation is structured to acquire GPS data, process it efficiently, and present output on an LCD, with additional features aimed at improving usability.

System Configuration and Startup

The process begins with both hardware configuration and firmware development. The microcontroller is programmed using Keil µVision, and communication interfaces are prepared for data transmission. The main initialization steps include:

- Uploading Embedded C firmware to the TM4C123G Launchpad via Keil.
- Configuring GPIO pins to establish connections with the GPS module, LCD, and TTL converter.
- Setting up UART to handle GPS data input and interpretation.

Real-Time GPS Signal Processing

The system continuously acquires GPS information via UART, enabling reliable real-time tracking. Each message includes location coordinates with timestamps, which the microcontroller processes seamlessly. Key operations include:

- Continuously reading GPS data from the UART buffer as it is received.
- Parsing NMEA protocol messages to extract latitude and longitude values.
- Validating signal integrity to ensure only accurate GPS data is used.

Data Processing & Display

Received coordinates are compared against a set of predefined locations stored in memory to identify the closest landmark. The system also performs the following tasks:

- Computes user speed by analyzing changes in position over time.
- Dynamically calculates the total distance traveled.
- Updates the LCD in real time with the name of the nearest identified landmark to support user navigation.

For improved accuracy especially around large landmarks multiple coordinate points are stored for each location. The following table illustrates examples of how GPS positions are mapped to specific faculty areas.

Location	Latitude	Longitude
Fountain	30.06564522	-31.27839088
Civil	30.06437302	-31.27773094
Louban Workshop	30.06307030	-31.27904892
Hall A	30.06445885	-31.28033829
Library	30.06504440	-31.27994537

System Validation and Evaluation

To ensure the system performed as intended, it was tested across multiple locations within the faculty premises. The validation process involved:

- Monitoring UART communication to confirm successful reception of GPS data.
- Performing live movement tests to verify that the displayed landmarks accurately matched real-world locations.
- Evaluating speed output by comparing calculated values with actual user movement in real-time conditions.

System Implementation and Technical Obstacles

Developing the system involved overcoming several hurdles before achieving stable and reliable performance. One of the initial obstacles was configuring UART communication between the microcontroller and the GPS module. At times, the module failed to transmit data correctly, requiring adjustments to baud rate and communication settings. Interpreting GPS messages delivered in raw NMEA format posed another difficulty, as the system had to isolate and clean meaningful data from unstructured text. Calculating traveled distance also demanded precise mathematical logic to maintain accuracy throughout the process.

Embedded Hardware-Software Coordination

Successfully connecting and coordinating the GPS module, LCD screen, and microcontroller required careful attention to both hardware setup and software initialization. Establishing stable UART communication was especially critical, as inconsistent signals occasionally caused incomplete data reception. In addition, the LCD had to be properly wired and formatted to present real-time information accurately without display glitches.

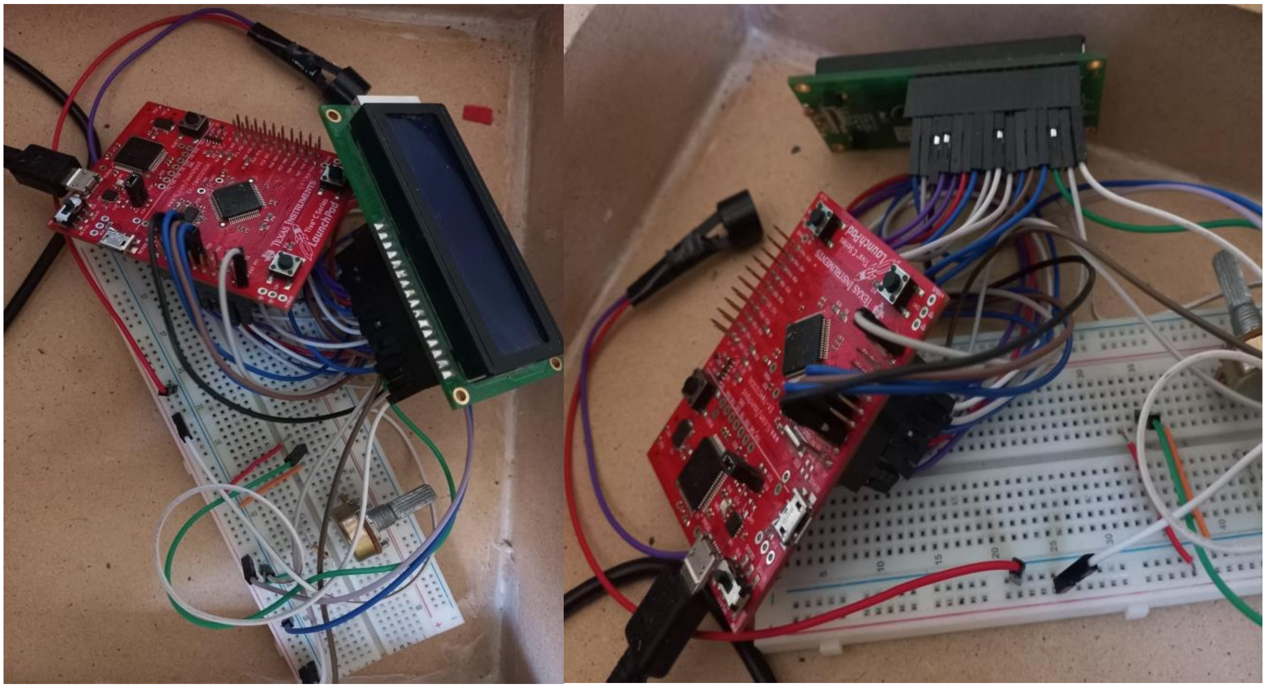

Figure 6: Real hardware connections

GPS Data Parsing

The incoming GPS data consisted of multiple NMEA sentences, only some of which contained useful position information. The microcontroller needed to filter out irrelevant lines and isolate the latitude and longitude values. Early attempts led to incorrect readings due to misinterpretation of the data stream, prompting the addition of validation checks. To handle scenarios with a weak or missing signal, fallback logic was implemented to avoid acting on invalid or incomplete data.

Distance Calculation

To measure the distance between the user's location and predefined landmarks, the system employed the Haversine formula. Unlike basic linear calculations, this method accounts for the Earth's curvature, providing an accurate estimate of the shortest path between two geographic points. This enabled realistic distance tracking across the campus. Additionally, a reset feature was implemented, allowing users to reinitialize distance tracking dynamically with the push of a button, ideal for restarting a session without rebooting the device.

Results & System Performance Review

Testing confirmed that the system effectively tracked user movement within the faculty grounds. Although occasional delays in GPS signal reception were observed, the system consistently identified nearby landmarks and displayed them on the LCD. The distance tracking feature functioned as intended, providing continuous updates during motion. When the reset button was pressed, the recorded distance was correctly cleared, allowing users to restart tracking seamlessly. Overall, the system delivered reliable performance for basic navigation tasks.

System Efficiency and Accuracy

To assess reliability, the system was evaluated under different signal conditions across multiple locations. It consistently acquired valid latitude and longitude coordinates, and its response was stable, even when GPS signal strength varied.

Live Demo

Real-time testing involved walking between known faculty landmarks to validate system accuracy. The microcontroller successfully matched stored coordinate data with physical locations and displayed the corresponding landmark names on the LCD without noticeable delay.

Figure 7: GPS tracking showing different locations.

Conclusion

This project showcases how a GPS-enabled embedded system can provide effective location tracking and landmark identification using a microcontroller. By parsing GPS data, referencing stored coordinates, and updating an LCD with the nearest location, the system delivers an interactive and practical tracking solution. Enhanced with features such as distance measurement, speed calculation, and buzzer-based signal feedback, the design improves user experience and functionality. UART communication plays a central role in ensuring smooth and continuous GPS data exchange. Testing verified that the system performs well in real-world scenarios within the faculty, offering precise location updates. Future development could explore improvements in graphical map display, integration of environmental sensors, or refined algorithms for greater accuracy. In its current form, the system is efficient, responsive, and well-suited for structured navigation tasks in embedded applications.

References

Texas Instruments Tiva C Series TM4C123GH6PM Datasheet

Project Live Demo Video

https://drive.google.com/file/d/1rp2W1kAiAGMmZB0BGwQVSX0d4spaZuYa/view

GitHub Repository

https://github.com/Ammar-Wahidi/GPS-Navigation-System